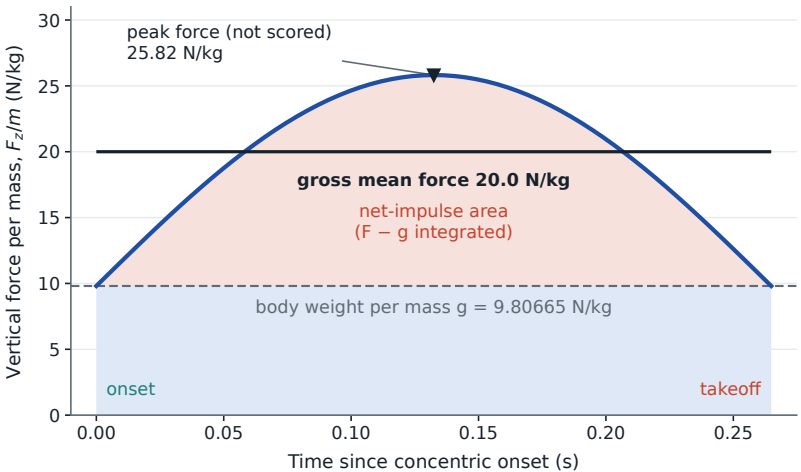
a **One simulated concentric trial (analytical half-sine)**



b **Three distinct quantities**

Gross mean concentric force

$$\bar{F} = \frac{1}{mT}\int_0^T F_z\,dt$$

scored force target

20.00 N/kg

Peak force

$$\max F_z/m$$

not a scored target

25.82 N/kg

Net concentric impulse

$$J = \frac{1}{m}\int_0^T (F_z - mg)\,dt$$

scored impulse target

2.70 m/s

Illustrative: $\bar{F}$ = 20 N/kg, J = 2.7 m/s ⇒ T = J/($\bar{F}$ − g) = 0.265 s. Peak = g + (π/2)($\bar{F}$ − g) for this half-sine. Not athlete data.